

1 Blowing Up $\mathbb{P}_K^3$ at Six Lines

The standard cubo-cubic transformation is given by blowing up a curve of arithmetic genus 3 in $\mathbb{P}^2$. Usually, this is taken to be a smooth curve, but we can also work with six lines in $\mathbb{P}^3$ arranged in either a tetrahedral formation or (as we shall investigate) four general lines and their two transversals.

Let us blow up $\mathbb{P}^3$ at four general lines $L_1, \dots, L_4$ and the two transversal lines T_1, T_2 . Call this blowup X . Then

$$\text{Pic}(X) = \langle H, L_1, L_2, L_3, L_4, T_1, T_2 \rangle$$

provides a basis in which we can express the cubo-cubic transformation Γ on X . Namely, we have

$$\Gamma = \begin{pmatrix} 3 & 2 & 2 & 2 & 2 & 0 & 0 \\ -1 & 0 & -1 & -1 & -1 & 0 & 0 \\ -1 & -1 & 0 & -1 & -1 & 0 & 0 \\ -1 & -1 & -1 & 0 & -1 & 0 & 0 \\ -1 & -1 & -1 & -1 & 0 & 0 & 0 \\ -1 & -1 & -1 & -1 & -1 & 1 & 0 \\ -1 & -1 & -1 & -1 & -1 & 0 & 1 \end{pmatrix}.$$

To note some particular examples of the behavior of Γ , observe that the class of a general plane H is transformed to a cubic surface containing the four skew lines (and thus necessarily contains the two transversals). The class of an exceptional locus L_i is transformed to a quadric surface containing the other three lines (and thus necessarily the two transversals as well). Each transversal is fixed by Γ . Of special use to us will be the transformation

$$\Gamma(H - L_i) = H - L_i \tag{1}$$

for each L_i . This identity will be relevant in constructing explicit formulae for Γ in the next section. For more information, refer to Unexpected surfaces singular on lines in $\mathbb{P}^3$ by Dumnicki, Harbourne, Roé, Szemberg, and Tutaj-Gasińska.

2 Explicit Construction

2.1 General Lines

Let $L_1, \dots, L_4$ be four general lines in $\mathbb{P}_K^3$, where K is algebraically closed. We shall construct the cubo-cubic transformation induced by these four lines.

Naïvely, the cubo-cubic transformation is simply the rational map induced by the linear system $\mathfrak{d} = |3H - L_1 - \dots - L_4|$ (note that we have unassigned base points as every cubic containing the four lines must also contain the two transversal lines thereof). However, finding an appropriate ordered basis of $\mathcal{L}(\mathfrak{d})$ is imperative, as this is how we guarantee that our transformation is an involution.

To begin, let

$$\mathcal{L}(H - L_i) = \langle p_i, p'_i \rangle$$

be polynomial generators for the linear system of planes surfaces containing L_i , and let

$$\mathcal{L}(2H - L_1 - L_2 - L_3 - L_4 + L_i) = \langle q_i \rangle$$

be the (unique) quadric surface containing all lines except L_i . Then we may start with the basis

$$\mathcal{L}(\mathfrak{d}) = \langle p_1 q_1, p'_1 q_1, p_2 q_2, p'_2 q_2 \rangle \cong K^4.$$

An important guiding feature of the cubo-cubic transformation Γ is that we want the transformations of p_i and p'_i to be some linear combination of $p_i q_i$ and $p'_i q_i$, which matches the action of the cubo-cubic matrix on the Picard group (see Identity (1) of the previous section). That is, we want

$$\Gamma(p_i), \Gamma(p'_i) \in \langle p_i q_i, p'_i q_i \rangle.$$

To this end, we want to find a linear transformation from $\langle p_1, p'_1, p_2, p'_2 \rangle = \mathcal{L}(H) \cong K^4$ to $\langle p_1 q_1, p'_1 q_1, p_2 q_2, p'_2 q_2 \rangle = \mathcal{L}(\mathfrak{d}) \cong K^4$ that transforms the following four planes in the prescribed manner:

- $\langle p_1, p'_1 \rangle \mapsto \langle p_1 q_1, p'_1 q_1 \rangle,$
- $\langle p_2, p'_2 \rangle \mapsto \langle p_2 q_2, p'_2 q_2 \rangle,$
- $\langle p_3, p'_3 \rangle = \langle A_3 p_1 + A'_3 p'_1 + B_3 p_2 + B'_3 p'_2, C_3 p_1 + C'_3 p'_1 + D_3 p_2 + D'_3 p'_2 \rangle \mapsto \langle p_3 q_3, p'_3 q_3 \rangle = \langle E_3 p_1 q_1 + E'_3 p'_1 q_1 + F_3 p_2 q_2 + F'_3 p'_2 q_2, G_3 p_1 q_1 + G'_3 p'_1 q_1 + H_3 p_2 q_2 + H'_3 p'_2 q_2 \rangle,$
- $\langle p_4, p'_4 \rangle = \langle A_4 p_1 + A'_4 p'_1 + B_4 p_2 + B'_4 p'_2, C_4 p_1 + C'_4 p'_1 + D_4 p_2 + D'_4 p'_2 \rangle \mapsto \langle p_4 q_4, p'_4 q_4 \rangle = \langle E_4 p_1 q_1 + E'_4 p'_1 q_1 + F_4 p_2 q_2 + F'_4 p'_2 q_2, G_4 p_1 q_1 + G'_4 p'_1 q_1 + H_4 p_2 q_2 + H'_4 p'_2 q_2 \rangle.$

In terms of vectors, our desired linear transformation exhibits the following properties:

- $\left\langle \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right\rangle \mapsto \left\langle \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right\rangle,$
- $\left\langle \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\rangle \mapsto \left\langle \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\rangle,$
- $\left\langle \begin{pmatrix} A_3 \\ A'_3 \\ B_3 \\ B'_3 \end{pmatrix}, \begin{pmatrix} C_3 \\ C'_3 \\ D_3 \\ D'_3 \end{pmatrix} \right\rangle \mapsto \left\langle \begin{pmatrix} E_3 \\ E'_3 \\ F_3 \\ F'_3 \end{pmatrix}, \begin{pmatrix} G_3 \\ G'_3 \\ H_3 \\ H'_3 \end{pmatrix} \right\rangle,$
- $\left\langle \begin{pmatrix} A_4 \\ A'_4 \\ B_4 \\ B'_4 \end{pmatrix}, \begin{pmatrix} C_4 \\ C'_4 \\ D_4 \\ D'_4 \end{pmatrix} \right\rangle \mapsto \left\langle \begin{pmatrix} E_4 \\ E'_4 \\ F_4 \\ F'_4 \end{pmatrix}, \begin{pmatrix} G_4 \\ G'_4 \\ H_4 \\ H'_4 \end{pmatrix} \right\rangle.$

2.2 Normal Form

Let us now arrange the four lines in what we will call normal form. We have the following four lines.

- $L_\infty = \text{rowspan} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$
- $L_0 = \text{rowspan} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix},$
- $L_1 = \text{rowspan} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix},$
- $L_2 = \text{rowspan} \begin{pmatrix} 1 & 0 & a & b \\ 0 & 1 & c & d \end{pmatrix},$

where $\begin{pmatrix} a & b \\ c & d \end{pmatrix}, \begin{pmatrix} a-1 & b \\ c & d-1 \end{pmatrix} \in \text{GL}(2, K)$. Every set of four mutually-skew lines can be assumed to exhibit this form. Then $\mathcal{L}(H - L_\infty) = \langle x, y \rangle$ and $\mathcal{L}(H - L_0) = \langle z, w \rangle$, so we will use $\langle x, y, z, w \rangle$ as our basis for $\mathcal{L}(H)$. Also note that $\mathcal{L}(H - L_1) = \langle x - z, y - w \rangle$ and $\mathcal{L}(H - L_2) = \langle (ad - bc)x - dz + cw, (ad - bc)y + bz - aw \rangle$.

Furthermore,

$$\mathcal{L}(2H - L_0 - L_1 - L_2) = \langle bxz + (bc - ad + d)yz - bz^2 + (-bc + ad - a)xw - cyw + (a - d)zw + cw^2 \rangle$$

and

$$\mathcal{L}(2H - L_\infty - L_1 - L_2) = \langle bx^2 + (-a + d)xy - cy^2 - bxz + (-d + 1)yz + (a - 1)xw + cyw \rangle,$$

which we will denote q_∞ and q_0 , respectively. Then we have

$$\mathcal{L}(\mathfrak{d}) = \langle xq_\infty, yq_\infty, zq_0, wq_0 \rangle$$

as our basis for cubics containing the four lines. Furthermore, note

$$\mathcal{L}(2H - L_\infty - L_0 - L_2) = \langle bxz + dyz - axw - cyw = q_1 \rangle$$

and

$$\mathcal{L}(2H - L_\infty - L_0 - L_1) = \langle xw - yz = q_2 \rangle.$$

Now we must write $(x - z)q_1$, $(y - w)q_1$, $((ad - bc)x - dz + cw)q_2$, and $((ad - bc)y + bz - aw)q_2$

as linear combinations of xq_∞ , yq_∞ , zq_0 , and wq_0 . We can write each as follows:

$$\begin{aligned} (x - z)q_1 &= \frac{\begin{pmatrix} a-1 & c & bc-ad+d & -c \end{pmatrix}}{bc-ad+a+d-1} \begin{pmatrix} xq_\infty \\ yq_\infty \\ zq_0 \\ wq_0 \end{pmatrix}, \\ (y - w)q_1 &= \frac{\begin{pmatrix} b & d-1 & -b & bc-ad+a \end{pmatrix}}{bc-ad+a+d-1} \begin{pmatrix} xq_\infty \\ yq_\infty \\ zq_0 \\ wq_0 \end{pmatrix}, \\ ((ad-bc)x - dz + cw)q_2 &= \frac{\begin{pmatrix} bc-ad+d & -c & -bc+ad-d & c \end{pmatrix}}{bc-ad+a+d-1} \begin{pmatrix} xq_\infty \\ yq_\infty \\ zq_0 \\ wq_0 \end{pmatrix}, \\ ((ad-bc)y + bz - aw)q_2 &= \frac{\begin{pmatrix} -b & bc-ad+a & b & -bc+ad-a \end{pmatrix}}{bc-ad+a+d-1} \begin{pmatrix} xq_\infty \\ yq_\infty \\ zq_0 \\ wq_0 \end{pmatrix}. \end{aligned}$$

Clearing out the denominator, our desired transformation of K^4 exhibits the following action on these four planar subspaces:

$$\begin{aligned} \bullet \quad & \left\langle \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right\rangle \mapsto \left\langle \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right\rangle, \\ \bullet \quad & \left\langle \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\rangle \mapsto \left\langle \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\rangle, \\ \bullet \quad & \left\langle \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix} \right\rangle \mapsto \left\langle \begin{pmatrix} a-1 \\ c \\ bc-ad+d \\ -c \end{pmatrix}, \begin{pmatrix} b \\ d-1 \\ -b \\ bc-ad+a \end{pmatrix} \right\rangle, \\ \bullet \quad & \left\langle \begin{pmatrix} ad-bc \\ 0 \\ -d \\ c \end{pmatrix}, \begin{pmatrix} 0 \\ ad-bc \\ b \\ -a \end{pmatrix} \right\rangle \mapsto \left\langle \begin{pmatrix} bc-ad+d \\ -c \\ -bc+ad-d \\ c \end{pmatrix}, \begin{pmatrix} -b \\ bc-ad+a \\ b \\ -bc+ad-a \end{pmatrix} \right\rangle. \end{aligned}$$

To find such a linear transformation on K^4 , let us perform the following change of basis. Let

$$\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} ad - bc \\ 0 \\ -d \\ c \end{pmatrix}, \begin{pmatrix} 0 \\ ad - bc \\ b \\ -a \end{pmatrix} \right\}.$$

Then under this new basis, we have the vectors

$$\begin{pmatrix} a - 1 \\ c \\ bc - ad + d \\ -c \end{pmatrix}, \begin{pmatrix} b \\ d - 1 \\ -b \\ bc - ad + a \end{pmatrix}, \begin{pmatrix} bc - ad + d \\ -c \\ -bc + ad - d \\ c \end{pmatrix}, \begin{pmatrix} -b \\ bc - ad + a \\ b \\ -bc + ad - a \end{pmatrix}$$

become

$$\begin{pmatrix} -bc + ad - 1 \\ 0 \\ \frac{-bc + ad - 1}{bc - ad} \\ \frac{-c}{bc - ad} \end{pmatrix}, \begin{pmatrix} 0 \\ -bc + ad - 1 \\ \frac{-b}{bc - ad} \\ \frac{-bc + ad - d}{bc - ad} \end{pmatrix}, \begin{pmatrix} bc - ad + d \\ -c \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -b \\ bc - ad + a \\ 0 \\ 0 \end{pmatrix},$$

respectively. Once again, we may clear our those pesky denominators and construct the following matrix. For space, we shall denote the determinant $ad - bc$ as Δ .

$$M = \begin{pmatrix} (\Delta^2 + 1)\lambda_1 & (\Delta^2 + 1)\lambda_3 & (-\Delta + d)\lambda_5 - b\lambda_6 & (-\Delta + d)\lambda_7 - b\lambda_8 \\ (\Delta^2 + 1)\lambda_2 & (\Delta^2 + 1)\lambda_4 & -c\lambda_5 + (-\Delta + a)\lambda_6 & -c\lambda_7 + (-\Delta + a)\lambda_8 \\ (-\Delta - 1)\lambda_1 - b\lambda_2 & (-\Delta - 1)\lambda_3 - b\lambda_4 & 0 & 0 \\ -c\lambda_1 + (-\Delta - d)\lambda_2 & -c\lambda_3 + (-\Delta - d)\lambda_4 & 0 & 0 \end{pmatrix}.$$

We want to find which values of λ_i cause the product

$$[I_4]_{\mathcal{E} \leftarrow \mathcal{B}} M [I_4]_{\mathcal{B} \leftarrow \mathcal{E}}$$

to be a block diagonal matrix of two 2×2 matrices. Solving for $\vec{\lambda}$, we end up finding $\vec{\lambda}$ is in the kernel of a rank 6 matrix, whose kernel is

$$\Lambda = \left\langle \begin{pmatrix} \frac{c}{bc - ad} \\ \frac{-a}{bc - ad} \\ \frac{b^2 c - a b d}{b^2 c - a b d} \\ \frac{bc - ad}{bc - ad} \\ \frac{-c}{bc - ad} \\ 0 \\ \frac{-c}{b} \\ 1 \\ 0 \end{pmatrix} = \psi_1, \begin{pmatrix} \frac{-a}{bc - ad} \\ \frac{a^2 + b c - a d}{b^2 c - a b d} \\ \frac{-b}{bc - ad} \\ \frac{a}{bc - ad} \\ \frac{-1}{bc - ad} \\ \frac{a - d}{b} \\ 0 \\ 1 \end{pmatrix} = \psi_2 \right\rangle.$$

Choosing $\psi = \mu_1 \psi_1 + \mu_2 \psi_2 \in \Lambda$, we get the block diagonal matrix

$$[I_4]_{\mathcal{E} \leftarrow \mathcal{B}} M [I_4]_{\mathcal{B} \leftarrow \mathcal{E}} = \begin{pmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{B} \end{pmatrix},$$

where

$$\mathbf{A} = \begin{pmatrix} \frac{-c}{bc-ad}\mu_1 + \frac{bc-ad+a}{bc-ad}\mu_2 & \frac{-bc+ad-d}{bc-ad}\mu_1 + \frac{b}{bc-ad}\mu_2 \\ \frac{bc^2-acd+ac}{b^2c-abd}\mu_1 + \frac{-abc+a^2d+bc-d-a^2d^2-a^2-bc+ad}{b^2c-abd}\mu_2 & \frac{c}{bc-ad}\mu_1 + \frac{-bc+ad-a}{bc-ad}\mu_2 \end{pmatrix}$$

and

$$\mathbf{B} = \begin{pmatrix} \frac{-bc^2+ac-d-a-c-d}{bc-ad}\mu_1 + \frac{abc-a^2d+a^2+bc}{bc-ad}\mu_2 & \frac{-bcd+ad^2-bc-d^2}{bc-ad}\mu_1 + \frac{b^2c-abd+ab+bd}{bc-ad}\mu_2 \\ \frac{abc^2-a^2cd+a^2c+bc^2}{b^2c-abd}\mu_1 + \frac{-a^2bc-b^2c^2+a^3d+2abc-d-a^2d^2-a^3-2abc+a^2d}{b^2c-abd}\mu_2 & \frac{bc^2-acd+ac+c+d}{bc-ad}\mu_1 + \frac{-abc+a^2d-a^2-bc}{bc-ad}\mu_2 \end{pmatrix}.$$

Since any choice of μ_1 and μ_2 is sufficient, we may be kind to ourselves by choosing nice coefficients; by choosing $\mu_1 = ab$ and $\mu_2 = bc$, we get a nice and simple matrix

$$[I_4]_{\mathcal{E} \leftarrow \mathcal{B}} M [I_4]_{\mathcal{B} \leftarrow \mathcal{E}} = \begin{pmatrix} bc & -ab+b & 0 & 0 \\ cd-c & -bc & 0 & 0 \\ 0 & 0 & bc & b^2c-abd+bd \\ 0 & 0 & -bc^2+acd-ac & -bc \end{pmatrix}$$

From this matrix we define the following four cubic polynomials.

- $\gamma_0 = bc(xq_\infty) + (cd-c)(yq_\infty)$,
- $\gamma_1 = (-ab+b)(xq_\infty) - bc(yq_\infty)$,
- $\gamma_2 = bc(zq_0) + (-bc^2+acd-ac)(wq_0)$,
- $\gamma_3 = (b^2c-abd+bd)(zq_0) - bc(wq_0)$.

Then $\langle \gamma_0, \gamma_1, \gamma_2, \gamma_3 \rangle = \mathcal{L}(3H - L_\infty - L_0 - L_1 - L_2)$ is an ordered basis that we may use to perform the cubo-cubic transformation

$$\Gamma(x : y : z : w) = (\gamma_0(x, y, z, w) : \gamma_1(x, y, z, w) : \gamma_2(x, y, z, w) : \gamma_3(x, y, z, w)).$$

Note: choosing $(\mu_1 : \mu_2) = (a-d \pm \sqrt{(a-d)^2 + 4bc} : 2c)$ seems to force $[I_4]_{\mathcal{E} \leftarrow \mathcal{B}} M [I_4]_{\mathcal{B} \leftarrow \mathcal{E}}$ to be an involution (in $\text{PGL}(4, K)$). However, this does not appear to be necessary for Γ to be an involution.

Remark 1. Fix the lines L_∞ , L_0 , and $L_1 = (I_2 \ I_2)$ and consider the matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, inducing the line $L_A = (I_2 \ A)$ such that $L_A \cap L_0 = L_A \cap L_1 = \emptyset$. Then the four lines $\{L_\infty, L_0, L_1, L_A\}$ induce a unique transversal if and only if A is not diagonalizable. Furthermore, the unique transversal line is given by $\begin{pmatrix} \mathbf{v} & 0 & 0 \\ 0 & 0 & \mathbf{v} \end{pmatrix}$, where $\mathbf{v}$ is a left-eigenvector of A ; when A is not diagonalizable, this is equivalent to $\begin{pmatrix} a-d & 2b & 0 & 0 \\ 0 & 0 & a-d & 2b \end{pmatrix}$.

3 Examples

Example 1. Let us choose the four skew lines in normal form given by $\{L_\infty, L_0, L_1, L_2\}$, where

$$L_2 = \text{rowspan} \begin{pmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 \end{pmatrix}.$$

Then the group of the groupoid is $C_3 \cong \left\langle \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix} \right\rangle \leq \text{PGL}(2, K)$. We have

$$q_\infty = xz + yz - z^2 + yw - zw - w^2,$$

$$q_0 = x^2 + xy + y^2 - xz - yz - yw$$

and so our four generators of the cubo-cubic transform centered at our four lines are

$$\gamma_0 = -(x + y)q_\infty = -x^2z - 2xyz - y^2z + xz^2 + yz^2 - xyw - y^2w + xzw + yzw + xw^2 + yw^2$$

$$\gamma_1 = yq_\infty = xyz + y^2z - yz^2 + y^2w - yzw - yw^2$$

$$\gamma_2 = -zq_0 = -x^2z - xyz - y^2z + xz^2 + yz^2 + yzw$$

$$\gamma_3 = (z + w)q_0 = x^2z + xyz + y^2z - xz^2 - yz^2 + x^2w + xyw + y^2w - xzw - 2yzw - yw^2.$$

Now let us consider the point $p = (0 : 0 : 1 : 0) \in L_\infty$. The orbit of p , denoted $[p]$, is the set of the following twelve points.

$$\begin{array}{lll} (0 : 0 : 1 : 0) & (0 : 0 : 0 : 1) & (0 : 0 : 1 : -1) \\ (1 : 0 : 0 : 0) & (0 : 1 : 0 : 0) & (1 : -1 : 0 : 0) \\ (1 : 0 : 1 : 0) & (0 : 1 : 0 : 1) & (1 : -1 : 1 : -1) \\ (1 : 0 : 1 : -1) & (0 : 1 : 1 : 0) & (1 : -1 : 0 : -1) \end{array}$$

Since the fibers of these points are lines in the exceptional loci of the four skew lines, their cubo-cubic transforms are lines as well, as given by the Picard group. Specifically, we have the images

$$\begin{array}{lll} \begin{pmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & 0 & 0 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix} \\ \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} & \begin{pmatrix} 1 & -1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix} \\ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix} & \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} & \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \\ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} & \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix} & \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{array}$$

respectively. Note that, though it may appear as though some of these lines intersect, in the blowup X they do not; all perceived intersection points are on the lines $\mathcal{L}$, and so they are actually separated on the exceptional loci. If we blow down the space after performing the cubo-cubic transformation, we attain a $(12_3, 12_3)$ -configuration of twelve lines and twelve triple points, as opposed to the original D_4 -configuration of $(12_4, 16_3)$, twelve points and sixteen lines.

Specifically, the twelve triple points are as follows.

$$\begin{array}{lll}
 (1 : 0 : 1 : -1) & (1 : -1 : 1 : -1) & (0 : 1 : 0 : 0) \\
 (1 : -1 : 0 : -1) & (0 : 1 : 0 : 1) & (1 : 0 : 0 : 0) \\
 (1 : 0 : 1 : 0) & (0 : 1 : 1 : 0) & (1 : -1 : 0 : 0) \\
 (0 : 0 : 0 : 1) & (0 : 0 : 1 : 0) & (0 : 0 : 1 : -1)
 \end{array}$$

Adding the four additional lines

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

yields a $(12_4, 16_3)$ -configuration of points and lines, isomorphic to D_4 . These are our original four lines $\mathcal{L}$!